Computer Engineering 4DK4
Lab 4
Simulation of ARQ Protocols

In this lab you will develop a discrete-event simulation to study the performance of three Automatic Repeat reQuest (ARQ) protocols: Stop-and-Wait (S&W), Go-Back-N (GBN), and Selective-Repeat (SR). Unlike the previous labs, no simulation program is supplied. You are responsible for designing and implementing the simulator. The simulated system consists of a single source transmitting packets to a single destination over one point-to-point link. You will consider both a continuously backlogged source and a source with a finite number of packets, and investigate the effects of window size, packet errors, timeout values, and ACK errors.

You will implement a general SR ARQ simulator with send window size W_s and receive window size W_r. By setting $W_r = 1$, the code can be used to simulate GBN ARQ, and by setting $W_s = W_r = 1$, the same code can be used to simulate S&W ARQ.

1 Preparation

1. The following default parameters will be used: fixed packet length of 1000 bits, constant link bit rate of 1 Mbps (and thus the transmission time of one data packet is 1 ms), the round-trip propagation delay of 10 ms (i.e., 5ms in each direction), and the timeout value of 15 ms. ACK transmission time and processing times are neglected.

2. Each data-packet transmission is independently unsuccessful with probability p_d. If a data packet is unsuccessful, it is discarded and no ACK is returned for that transmission. For successfully received packets, each ACK is independently unsuccessful with probability p_a. Assume that data-packet and ACK errors are independent.

3. Use multiple random number generator seeds for experiments involving errors. Include the student ID number of at least one group member as one of the seeds. Use the same seeds when comparing different ARQ protocols.

4. Make any additional reasonable assumptions required by your implementation and clearly state them in your report.

2 Experiments

1. **Simulation verification.** First set $p_d = p_a = 0$, and then set $p_d = 0.5$ and $p_a = 0.5$. For each combination of p_d and p_a, generate suitable short event traces to demonstrate that your

S&W, GBN and SR simulators respond correctly to data-packet and ACK errors. Describe how you verified the correctness of your simulator.

2. **Continuously backlogged source: without errors.** Assume that the source always has packets available for transmission. Set $p_d = p_a = 0$. For GBN and SR, vary W_s over a suitable range. Your range should include W_s values both below and above the value required to fully utilize the link. Generate a plot of link efficiency versus W_s, showing GBN and SR on the same graph. For SR, generate the efficiency-versus-W_s curves for at least 3 different W_r values. Include the S&W efficiency as a reference. Compare your simulation results with the efficiency performance expected from the analysis covered in lectures.

   Discuss the relationship between link efficiency and window sizes as well as the similarities and differences between GBN and SR under error-free conditions.

3. **Continuously backlogged source: with packet errors and without ACK errors.** Continue to assume a continuously backlogged source. Set $p_a = 0$, and consider non-zero data-packet error probabilities.

   (a) For GBN and SR, select W_s that is sufficiently large so that the link can operate at its maximum efficiency when it is error free. Vary p_d over a suitable range, including $p_d = 0$, and generate a plot of efficiency versus p_d for GBN and SR. For SR, generate the efficiency-versus-p_d curves for at least 3 different $W_r(> 1)$ values. Include the S&W efficiency as a reference. Compare your simulation results with the efficiency performance expected from the analysis covered in lectures.

   (b) Choose a suitable non-zero value of p_d and vary W_s for GBN and SR. Plot efficiency versus W_s for the two protocols. Also collect and plot the average number of data-packet transmission attempts per successfully delivered packet versus W_s. For SR, plot curves for 3 different $W_r(> 1)$ values. Discuss the differences between GBN and SR.

4. **Continuously backlogged source: with both packet and ACK errors.** Use the same continuously backlogged source as in the previous parts. Choose a suitable non-zero data-packet error probability and vary the ACK error probability over a range starting from $p_a = 0$. Use the same GBN and SR window sizes as in Part 3(a). Generate plots showing (i) link efficiency versus p_a; and (ii) average number of data-packet transmission attempts per successfully delivered packet versus p_a. Show S&W, GBN, and SR on the same graph for each performance measure. Select appropriate window sizes for GBN and SR to demonstrate their effects.

   Compare the effects of packet errors and ACK errors on the three ARQ protocols. Explain any important differences you observe and how the window sizes are related to these differences.

5. **Effect of retransmission timeout.** Assume a continuously backlogged source. Choose suitable non-zero values of p_d and p_a, and use a fixed W_s for GBN and SR that is sufficiently large to provide high link utilization when there are no errors. Vary the retransmission timeout T_{out} over a suitable range. Your range should include timeout values shorter than, moderately longer than, and substantially longer than the round-trip time.

For S&W, GBN, and SR, generate plots of (i) link efficiency versus T_{out}, and (ii) average number of data-packet transmission attempts per successfully delivered packet versus T_{out}. Use the same error probabilities when comparing the three protocols. For SR, choose 3 different $W_r > 1$ values to generate the curves.

Discuss the following:

i) the effects of choosing a timeout that is much longer or shorter than necessary; and
ii) the different effects of the timeout value among S&W, GBN, and SR.

6. **Finite packet transfer.** The source now contains a finite number of packets at time $t = 0$. No additional packets arrive during the simulation. Define the transfer completion time as the time from the beginning of transmission of the first packet until the source has received acknowledgement that all packets have been successfully delivered. Use $N = 100$ packets initially, and vary the GBN and SR window sizes over a suitable range.

   Consider the following cases: ($p_d = p_a = 0$), ($p_d = 0.05$, $p_a = 0$), and ($p_d = p_a = 0.05$). For each case, generate a plot of mean transfer completion time versus W_s, showing GBN and SR on the same graph. For SR, plot curves for 3 different $W_r (> 1)$ values. Include the corresponding S&W result as a reference.

   Discuss:

   i) how the window sizes affect the completion time of a finite transfer;
   ii) how packet errors change the relationship between window sizes and completion time; and
   iii) how ACK errors further affect the results.

7. **Effect of transfer size.** Repeat a suitable subset of Part 6 using at least two additional values of N, one substantially smaller and one substantially larger than 100. Explain how the number of packets being transferred affects the relationship between file transfer completion time and ARQ window sizes.

3 Submission

Each group (maximum of three students) should submit a single writeup for the lab. The writeup should clearly document your simulation design, experiments, results, and observations.

1. Simulation design: Describe the major events and state variables used in your simulation. Explain how the ARQ protocols, retransmission timers, data-packet errors, and ACK errors are represented. Clearly state any assumptions or design choices.

2. Experimental setup: Clearly describe the parameter values used for each experiment, including window sizes, number of packets, error probabilities, simulation run lengths, and random-number generator seeds. Clearly identify which parameters were varied and which were held fixed.

3. Results: Include all plots requested in Section 2. Select appropriate parameter ranges and sufficient simulation data points so that the important operating regions and performance trends can be clearly observed.

4. Analysis and discussion: Explain the important observations from your results and the reasons for the observed behavior. Where comparisons are requested, clearly identify and explain the similarities and differences between the protocols and operating conditions.

5. Simulation verification and reliability: Describe how you verified the correctness of the three ARQ implementations. Use sufficiently long simulation runs and, where appropriate, multiple random-number generator seeds to obtain reliable results.

6. Code: Submit the complete simulation program separately. In the report, include only selected code or pseudocode where useful for explaining important implementation decisions.

7. Plots: Clearly label all axes, units, curves, and relevant parameter values. Use markers to identify simulation data points. Do not use piecewise-linear interpolation or overfitted curves to produce artificially smooth performance curves.

The report is marked according to the above submission requirements. The CEAB indicators below are provided for reference and show how selected components of the lab contribute to graduate attribute assessment.

Table 1: CEAB Assessment Rubric

Indicator	3 – Exceeds Expectations	2 – Meets Expectations	1 – Partially Meets Expectations	0 – Does Not Meet Expectations
2.1 Problem Analysis	Clearly identifies relevant assumptions and design choices and correctly applies appropriate ARQ and discrete-event simulation principles in developing the simulation model.	States reasonable assumptions and design choices and generally applies appropriate ARQ and simulation principles correctly.	Some assumptions or design choices are unclear, incomplete, or weakly connected to the engineering principles used in the simulation.	Important assumptions or design choices are missing or unreasonable, and/or the simulation is based on incorrect engineering principles.
2.2 Problem Analysis	Explains observed simulation behaviour using well-substantiated reasoning. Comparisons among protocols and operating conditions are clear and insightful, and relevant limitations of the results are recognized.	Provides reasonable explanations for the main observed trends and correctly discusses the important similarities and differences among protocols and operating conditions.	Provides mostly descriptive observations with limited supporting reasoning; important comparisons or limitations are only partially addressed.	Provides little or no meaningful analysis of the results, or explanations are inconsistent with the simulation results or underlying ARQ behaviour.
3.1 Investigation	Selects appropriate window sizes, error probabilities, timeout values, transfer sizes, run lengths, random seeds, and numbers of data points so that important operating regions and performance trends are clearly demonstrated; variability and reliability are appropriately considered.	Uses generally appropriate parameter ranges, run lengths, random seeds, and sufficient data points to support the main conclusions.	Some choices of parameter ranges, run lengths, random seeds, or numbers of data points are insufficient, making some trends or conclusions unclear or unreliable.	Simulation settings or data-collection methods are inappropriate or insufficient to support the reported conclusions.
4.4 Design	Clearly justifies the simulation architecture and implementation decisions, including events, state variables, window and timer handling, error modelling, and assumptions, with thoughtful consideration of relevant constraints and limitations.	Provides reasonable justification for the major simulation design and implementation decisions, with appropriate consideration of assumptions and constraints.	Design and implementation decisions are described but only weakly justified; relevant assumptions, constraints, or limitations are incompletely considered.	Design or implementation decisions are unexplained, inappropriate, or inconsistent with the required ARQ system model.